Instruction to Generate Nondegenerate ETPA Figures using Matlab code

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Instructions for using the Matlab codes to generate the nondegenerate ETPA figures (blue figures)

1. Place all the provided Matlab codes and the .mat file (output of transition_dipole.py) in the same directory.
2. In the run_etpa_sgrid_vs_igrid.m code
   - Change the variable in line 3 to match the name of the .mat file that you are using (without the extension)
   - Lines 5-18 are where the parameters that we enter are set.
     - Energies should be entered in eV
       - it is possible to set up for other units but the conversion factor I have set up later in code uses eV
     - Entanglement time should be in seconds
3. Run the run_etpa_sgrid_vs_igrid.m code
   - This will generate the nondegenerate ETPA figure
4. Due to the size of the figure, we cannot directly save the .fig file for the figure. So I have written a code to replot the figure from the saved data file.
5. Running the ETPA_2d_signolvidler.m code will take the sRgrid_i2_1-s5_5 (default name) file that was generated by run_etpa_sgrid_vs_igrid.m and remake the figure.
6. It may be necessary to adjust the range of data that you are plotting to make the color scale easier to read.